

- Hide menu
- Working with Medical Volumes
- Registering Medical Volumes
- Video: Introduction to Volume Registration2 min
- Reading: Key Functions in the Book Video5 min
- Video: Registering Medical Volumes5 min
- Reading: Practice Registering Medical Volumes15 min
- Graded Assignment: Graded Assignment: Viewing and Registering 3D Medical ImagesSubmitted
- Optional: Additional Resources for Registration

Your grade: 100%

Your latest: 100% • Your highest: 100% • 3 attempts • 100% of 100% (All items you did not attempt)

Next item →

Graded Assignment: Viewing and Registering 3D Medical Images

1. Use `medicalVolume` to load the data. How many sagittal slices are there? 1 / 1 point

96

COOCH

Ready to review what you've learned before starting the assignment? I'm here to help.

Correct

Help me practice

Let's chat

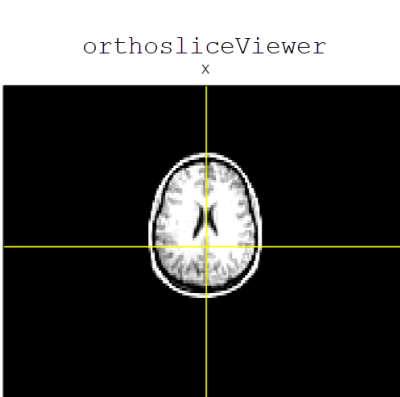
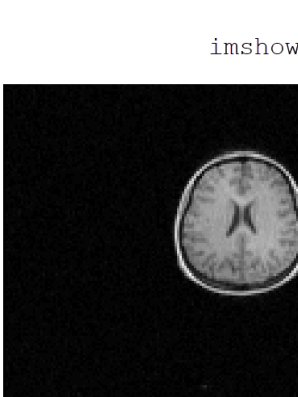
In addition to the `medicalVolume`, the `PlaneMapping` property indicates that the sagittal slices are stored in the 3rd dimension of the `VoxelData` array, which has a size of 96.

Assignment details

2. Compare the two images below. ~~Which one is the slice of the T1_3T_2018 MRI volume included in the course files. The image on the left was obtained using `extractSlice` and `imshow` while the image on the right was generated by `orthosliceViewer`.~~ 1 / 1 point

Submitted

Why does the scale of the images look different?



See feedback

☐ `imshow` scales medical images correctly and should be used.

☒ `orthosliceViewer` uses the metadata to properly scale the display in the x and y directions while `imshow` does not.

☐ `orthosliceViewer` uses the metadata to adjust the intensity display of the images.

Correct

Correct! `orthosliceViewer` use metadata to properly scale the display. The `imshow` function doesn't know about the metadata so uses the number of pixels in each direction.

3. You want to register two volumes, but the plane mapping, voxel spacing, and number of slices differ. To correct for this, you use `resample` to convert `volume1` into the same coordinate system as `volume2`. 1 / 1 point

Which line of code accomplishes this?

☐ `resampledVol = resample(volume2, volume1.VolumeGeometry)`

☐ `resampledVol = resample(volume1, volume2)`

☐ `resampledVol = resample(volume2, volume1)`

☒ `resampledVol = resample(volume1, volume2.VolumeGeometry)`

Correct

The first input is the medical volume you want to resample. The second is the `VolumeGeometry` of the target medical volume.

4. You have data from two MRI scans you want to register. One used a sequence that acquires T1-weighted images while the other used a sequence that acquires T2-weighted images. 1 / 1 point

What modality setting should you use in `IsaregionFig`?

☒ "multimodal"

☐ "monomodal"

Correct

Multimodal images have different brightness and contrast. The images can come from two different medical imaging modalities (like CT and MRI). The images can also come from a single device, such as an MRI scanner using different imaging sequences.

5. You want to register two volumes taking into account differences due to translation, rotation, and scale. Which transformation type should you use? 1 / 1 point

☐ "translation"

☐ "rigid"

☐ "affine"

☒ "similarity"

Correct

This option assumes the volumes differ by a global shift in (x,y,z), a global rotation, and a global scale factor.

Close ×



You're ahead of the game!
Continue this momentum and you'll finish
3 days earlier than expected.